

Equivalence Classes and Boundary Values

1. Equivalence classes

- income: $i \leq 1499$ (invalid), $1500 \leq i \leq 80K$ (valid), $i \geq 80001$ (invalid)
- #mortgage: $\#m \leq -1$ (invalid), $0 \leq \#m \leq 5$ (valid), $\#m \geq 6$ (invalid)
- applicant type: {Person} (valid), {Company, Trust, ...} (invalid)
- property type: {HDB, Cando} (valid), {TownHouse, MobileHome, ...} (invalid)

2. Boundary values

- income: 1499, 1500, 1501, 79999, 80000, 800001
- #mortgage: -1, 0, 1, 4, 5, 6
- applicant type: N/A
- property type: N/A

Valid values

Invalid values

Design Test Cases Using ECs and BVs

1. Test valid inputs of several parameters
 - Test valid inputs of several parameters at the same time
 - Choose one valid input from an equivalence class of each input parameter
2. Test invalid inputs of several parameters
 - Test **ONLY one invalid** input from an equivalence class of one input parameter per test case
 - Choose valid inputs from an equivalence class of all other input parameters

Use the boundary values of the equivalence classes of range-of-values parameters

- Value **on-the-boundary** as **valid** input
- **Just-below** or **just-above** values (if not in other equivalence classes) as **invalid** inputs

If the equivalence class is **small**, you may create test cases for **all the values** in the equivalence class (e.g., PropertyType)